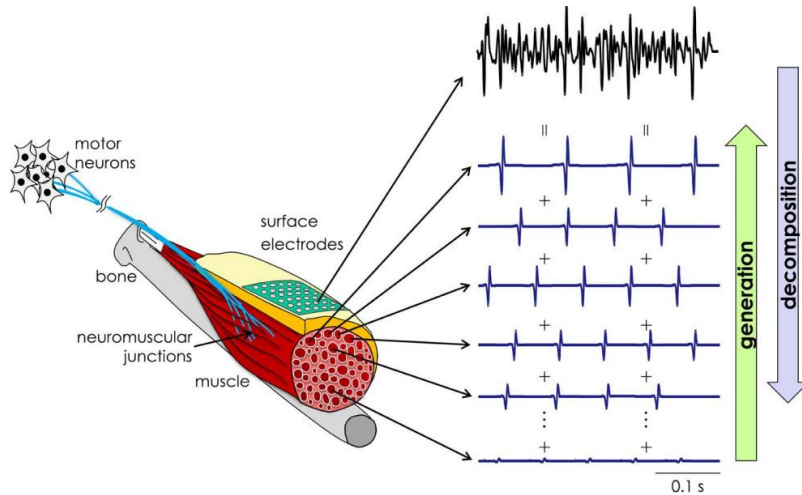


Introduction

ABOUT sEMG



"Characterization of Human Motor Units From Surface EMG Decomposition"

- **sEMG(Surface Electromyography)** signal is the algebraic summation of motor unit action potential (MUAP).
- Features : non-invasive(+), easy to get(+), a lot of noises(-)
- Applicable to prosthetic control, rehabilitation robotics, etc.

Limitation of Previous Studies

- Predominant use of **multi-channel, high-cost** EMG systems (e.g., Delsys Trigno)
- Existing single-channel studies are **limited to simple finger or wrist movements**, which are not sufficient for controlling diverse functional grasps in prosthetic hands
- Lack of systematic analysis on **electrode placement**

Research Objective

- Use a **low-cost, single-channel EMG sensor** (e.g., Myoware 2.0)
- Classify **four functional hand grasps**: power, pinch, lateral, tripod
- Propose **electrode placement guidelines**
- **Compare with dual-channel setups** to validate the efficacy and simplicity of single-channel systems



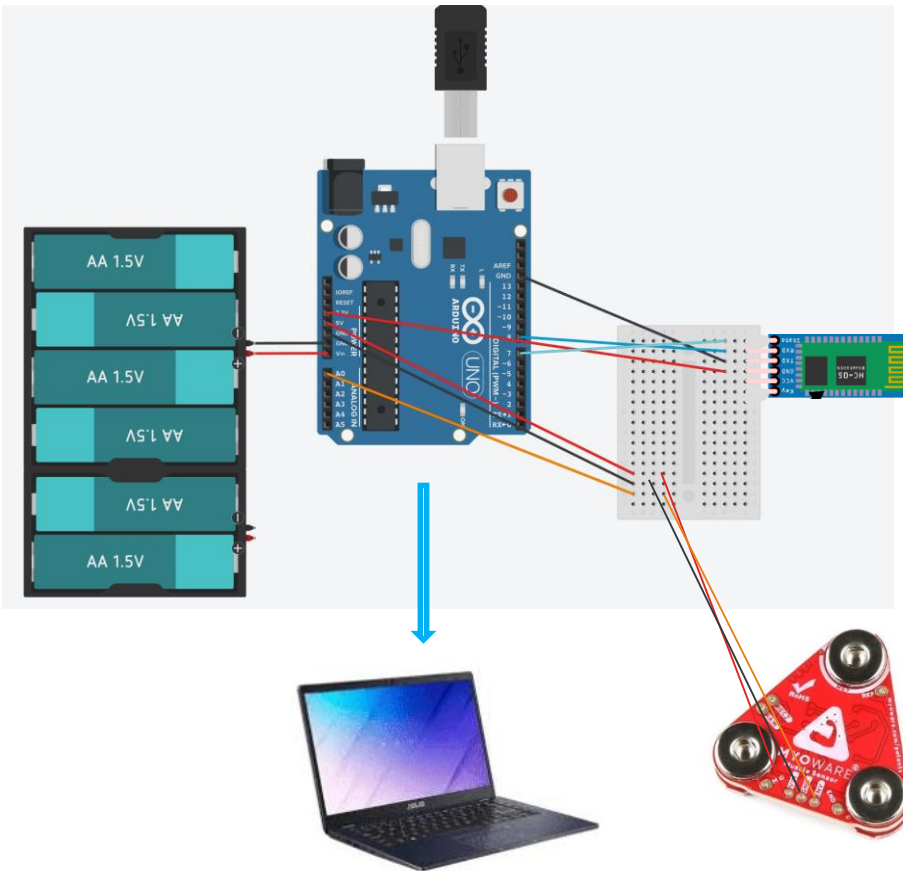
<Myoware 2.0>



<Delsys Trigno>

Test Bench & Experimental Setup

Test Bench Setup



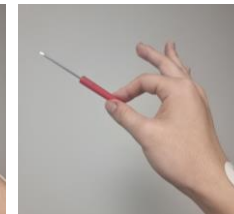
Arduino → Myoware V2 → Bluetooth → PC(python)

Experimental Protocol

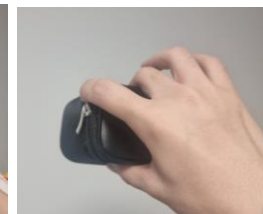
- Subject: Single participant (author)
- Grasping strategy : Power, Pinch, Tripod, Lateral grasp
- Procedure per set:
 - Perform grasping for 5 seconds & Relax for 5 seconds
 - 29 repetitions per each grasping
- To address variability (muscle fatigue, position shifts, etc.), data collection was repeated for **2 independent sets**
- The electrode was attached to **five different forearm muscles in separate sets** to compare classification performance across placements
- Total Samples:
 $4 \text{ grasps} \times 29 \text{ repetitions} \times 2 \text{ sets} \times 5 \text{ channels} =$
1,160 segments



**Power
grasp**



**Pinch
grasp**



**Tripod
grasp**



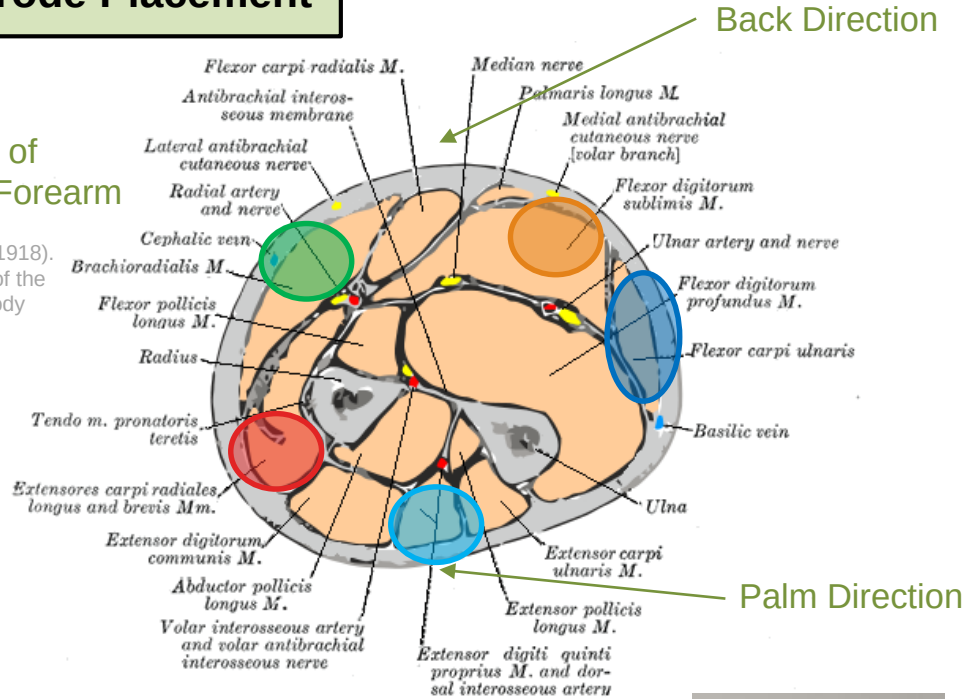
**Lateral
grasp**

Electrode Placement, Signal Example & Analysis Workflow

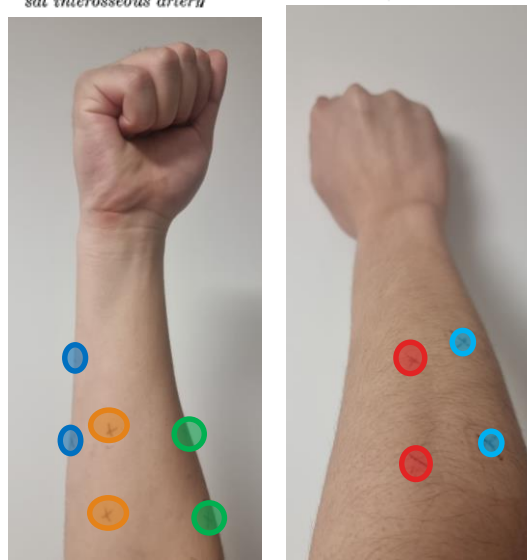
Electrode Placement

Figure of Right Forearm

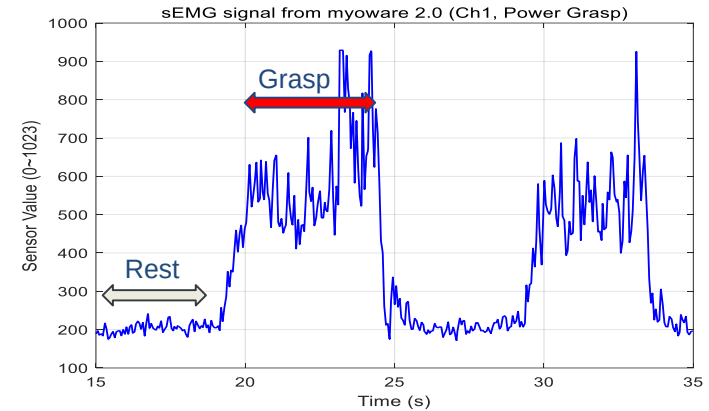
Gray, H. (1918).
Anatomy of the
Human Body



- Ch1 : Extensor Carpi Radialis (Red circle)
- Ch2 : Brachioradialis (Green Circle)
- Ch3 : Flexor digitorum sublimis (Orange circle)
- Ch4 : Flexor Carpi Ulnaris (Blue Circle)
- Ch5 : Extensor digitorum (SkyBlue Circle)



Signal Example



Analysis WorkFlow – Used Model

<SVM>

- Extract features(MAV, RSM, WL, SSC) from each channel

<1D CNN>

- Convolutional layers: (1 or 2 → 16) → (16 → 32), followed by global pooling

<GRU>

- Bidirectional GRU (2 layers, hidden size 64), last hidden state for classification.

→ All models were evaluated using **5-fold cross-validation** to ensure reliability.

Results & Conclusion

Experiment Result

- The experiment included **5 single-channel placements**, and **10 dual-channel combinations** by pairing them

- **Average accuracy :**

- Single-channel (5 channels):

SVM: 77.39%
1D-CNN: 75.97%
GRU: 75.20%

- Dual-channel (10 combinations):

SVM: 87.03%,
1D-CNN: 88.10%,
GRU: 87.81%

- **Best single-channel:**

- Ch4 (FCU), 87.03% (1D-CNN)

- **Top 3 dual-channel pairs :**

- Ch3+Ch4, Ch1+Ch4, Ch2+Ch4

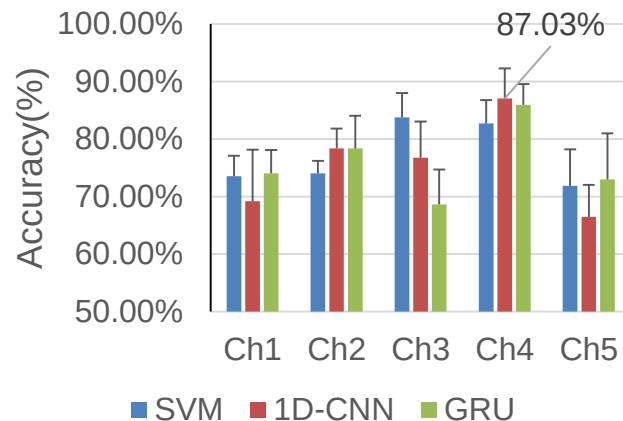
- **Bottom 3 dual-channel pairs:**

- Ch1+Ch2, Ch1+Ch3, Ch2+Ch3

- **Highest overall accuracy:**

- Ch3+Ch4, 95.14% (1D-CNN)

**Classification Accuracy
for Single-Channel
(5-Fold Cross-Validation)**



Conclusion

- Ch4 (FCU), which showed the best single-channel performance, **appeared in every top-performing dual-channel combination**, suggesting that this location provides robust, discriminative signals both alone and in combination.
- While **dual-channel setups averaged 87–88% accuracy**, the comparable performance of Ch4 alone highlights the **practical feasibility of simpler, single-sensor systems**.
- The **Ch3+Ch4 pair**, which achieved the highest accuracy (up to **95.14%**), consists of anatomically adjacent muscles — potentially acting like **an enriched, time-extended signal source**.
- It implies that **a single-source electrode—when placed optimally—can yield denser data**, enhancing the separability of grasp patterns even without additional channels.

**Classification Accuracy
for Top 3 and Bottom 3
Dual-Channel
(5-Fold Cross-Validation)**

